

Possible issue in pegasos classifier: Not training for data dimension besides

<https://github.com/qiskit-community/qiskit-machine-learning/issues/344>

ROW 82

Possible issue in pegasos classifier: Not training for data dimension besides 2 #344

New issue



Closed

#345



charmerDark opened on Mar 10, 2022

Environment

- Qiskit Machine Learning version: 0.3.1
- Python version: 3.7.12
- Operating system: Ubuntu 18.04

What is happening?

The `PegasosQSVC` method seems to have issues taking in data of any format except `(num_samples, 2)`. The fit method throws errors when using any other feature size. However the documentation mentions the arguments for `pegasus_qsvc.fit` being `(num_samples, num_fetaures)` and does not limit it to 2.

How can we reproduce the issue?

Taken from the [pegasus tutorial](#)

```
from sklearn.datasets import make_blobs
features, labels = make_blobs(n_samples=20, n_features=4, centers=2, random_state=3, shuffle=True)

import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler

train_features, test_features, train_labels, test_labels = train_test_split(features, labels, train_size=15,

num_qubits = 4
tau = 100
C = 1000

from qiskit import BasicAer
from qiskit.circuit.library import ZFeatureMap
from qiskit.utils import QuantumInstance, algorithm_globals

from qiskit_machine_learning.kernels import QuantumKernel

algorithm_globals.random_seed = 12345

pegasos_backend = QuantumInstance(
    BasicAer.get_backend("statevector_simulator"),
    seed_simulator=algorithm_globals.random_seed,
    seed_transpiler=algorithm_globals.random_seed,
)

feature_map = ZFeatureMap(feature_dimension=num_qubits, reps=1)
qkernel = QuantumKernel(feature_map=feature_map, quantum_instance=pegasos_backend)

from qiskit_machine_learning.algorithms import PegasosQSVC

pegasos_qsvc = PegasosQSVC(quantum_kernel=qkernel, C=C, num_steps=tau)
pegasos_qsvc.fit(train_features, train_labels)
```

yields the error

```
-----
ValueError                                Traceback (most recent call last)
<ipython-input-23-5efd7aedaf88> in <module>
      4
      5 # training
----> 6 pegasos_qsvc.fit(train_features, train_labels)
      7
      8 # testing
```

Assignees

No one assigned

Labels

type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Quantum kernel fix for dimensions other than 2
qiskit-community/qiskit-machine-learning

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants



```
~/local/lib/python3.7/site-packages/qiskit_machine_learning/algorithms/classifiers/pegasos_qsvc.py in fit(self, X, training=True)
184     value = self._compute_weighted_kernel_sum(i, X, training=True)
185
--> 186     if (self._label_map[y[i]] * self.C / step) * value < 1:
187         # only way for a component of alpha to become non zero
188         self._alphas[i] = self._alphas.get(i, 0) + 1

ValueError: The truth value of an array with more than one element is ambiguous. Use a.any() or a.all()
```

while substituting `n_features=4` with `n_features=7` and changing the corresponding `num_qubits` to 7 yields the error.

```
-----
ValueError                                Traceback (most recent call last)
<ipython-input-29-5efd7aedaf88> in <module>
      4
      5 # training
----> 6 pegasos_qsvc.fit(train_features, train_labels)
      7
      8 # testing

~/local/lib/python3.7/site-packages/qiskit_machine_learning/algorithms/classifiers/pegasos_qsvc.py in fit(self, X, training=True)
182     i = algorithm_globals.random.integers(0, len(y))
183
--> 184     value = self._compute_weighted_kernel_sum(i, X, training=True)
185
186     if (self._label_map[y[i]] * self.C / step) * value < 1:

~/local/lib/python3.7/site-packages/qiskit_machine_learning/algorithms/classifiers/pegasos_qsvc.py in _compute_weighted_kernel_sum(self, i, X, training=True)
266     # evaluate kernel function only for the fixed datum and the data with non zero alpha
267     kernel[(index, j)] = kernel.get(
--> 268         (index, j), self._quantum_kernel.evaluate(X[index], x_j)
269     )
270     kernel_value = kernel[(index, j)]

~/local/lib/python3.7/site-packages/qiskit_machine_learning/kernels/quantum_kernel.py in evaluate(self, x_vec, y_vec)
408
409     if x_vec.ndim == 1:
--> 410         x_vec = np.reshape(x_vec, (-1, 2))
411
412     if y_vec is not None and y_vec.ndim > 2:

<__array_function__ internals> in reshape(*args, **kwargs)

~/local/lib/python3.7/site-packages/numpy/core/fromnumeric.py in reshape(a, newshape, order)
296     [5, 6]])
297     """
--> 298     return _wrapfunc(a, 'reshape', newshape, order=order)
299
300

~/local/lib/python3.7/site-packages/numpy/core/fromnumeric.py in _wrapfunc(obj, method, *args, **kwargs)
55
56     try:
--> 57         return bound(*args, **kwargs)
58     except TypeError:
59         # A TypeError occurs if the object does have such a method in its

ValueError: cannot reshape array of size 7 into shape (2)
```

What should happen?

Expected output would be either a working `pegasusvc.fit` method or an updated documentation where the `num_dimensions` is shown as 2.

Any suggestions?

No response



Quantum kernel fix for dimensions other than 2 #345

< Code

Merged adekusal-drl merged 4 commits into `qiskit-community:main` from `gentinettagian:quantum_kernel_fix` on Mar 10, 2022

Conversation 5 Commits 4 Checks 0 Files changed 3

+38 -4



gentinettagian commented on Mar 10, 2022 · edited by woodsp-ibm

Contributor

Fixes #344

Summary

The `QuantumKernel.evaluate()` method was hardcoded to only work with 2-dimensional data, if the input vectors are 1D-arrays. This error arises in the `PegasusQSV` algorithm, where single kernel entries are computed once at a time.

Details and comments



gentinettagian added 2 commits 3 years ago

fixed quantum_kernel 315215f

Merge branch 'main' of <https://github.com/gentinettagian/qiskit-machi...> 4439ef2

gentinettagian requested review from stefan-woerner, pbark, manaelmarques, woodsp-ibm and adekusal-drl as code owners 3 years ago



adekusal-drl reviewed on Mar 10, 2022

View reviewed changes

adekusal-drl left a comment · edited

Collaborator

Can you please add a short test that covers this change and a reno file?



adekusal-drl added `stable backport potential` `Changelog: Bugfix` labels on Mar 10, 2022

gentinettagian force-pushed the quantum_kernel_fix branch from `0eff507` to `3721c0c` 3 years ago

Compare

year fixed f687128

gentinettagian force-pushed the quantum_kernel_fix branch from `3721c0c` to `f687128` 3 years ago

Compare



coveralls commented on Mar 10, 2022 · edited

...

Pull Request Test Coverage Report for Build 1964857883

- 2 of 2 (100.0%) changed or added relevant lines in 1 file are covered.
- No unchanged relevant lines lost coverage.
- Overall coverage remained the same at 85.97%

Reviewers

adekusal-drl	✓
stefan-woerner	●
pbark	●
manaelmarques	●
woodsp-ibm	●

Assignees

No one assigned

Labels

`Changelog: Bugfix` `stable backport potential`

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

✓ Possible issue in pegasus classifier: Not traini...

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

3 participants

